

InvestSmart Database Script and Setup Guide

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Project: InvestSmart Financial Management and Investment Recommendation System

Database engine: MySQL / MariaDB

Recommended local stack: XAMPP

Database name: investsmart

Purpose

This document contains the database setup information for the InvestSmart project. It is intended to help a lecturer, student, or developer recreate the database tables, relationships, seed data, and OTP update table used by the backend.

How to Run the Database Script in XAMPP

1. Start XAMPP Control Panel.
2. Start Apache and MySQL.
3. Open a browser and go to <http://localhost/phpmyadmin>.
4. Create a database named investsmart if it does not already exist.
5. Select the investsmart database.
6. Open the SQL tab.
7. Copy and run the SQL script from database/schema.sql.
8. If the database already existed before OTP was added, also run database/update_otp_table.sql.

Command Prompt Option

If MySQL is available from the command prompt, run:

```
cd C:\xampp\mysql\bin
mysql -u root
CREATE DATABASE IF NOT EXISTS investsmart;
USE investsmart;
SOURCE C:/xampp/htdocs/investsmart/database/schema.sql;
```

Important Tables

- users: stores login credentials, role, and account status.
- clients: stores normal user profile details linked to users.
- admins: stores administrator profile details linked to users.
- financial_profiles: stores salary, deductions, expenses, savings, and net salary.
- banks: stores investment provider/bank criteria used by the calculator.
- investment_plans: stores saved recommendations/plans for clients.
- activity_logs: stores audit trail records.
- auth_otps: stores hashed OTP records for login and password reset.

Default Seed Data

The schema creates the default administrator account and seed bank/provider records.

Database Script

```
CREATE DATABASE IF NOT EXISTS investsmart CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;
USE investsmart;
```

```
CREATE TABLE IF NOT EXISTS users (
  id INT AUTO_INCREMENT PRIMARY KEY,
  email VARCHAR(180) NOT NULL UNIQUE,
  password_hash VARCHAR(255) NOT NULL,
  role ENUM('client', 'admin') NOT NULL DEFAULT 'client',
  status ENUM('active', 'inactive') NOT NULL DEFAULT 'active',
```

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```
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

```
CREATE TABLE IF NOT EXISTS clients (
    id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT NOT NULL UNIQUE,
    full_name VARCHAR(120) NOT NULL,
    surname VARCHAR(120) NOT NULL,
    id_number VARCHAR(20) NOT NULL UNIQUE,
    contact_info VARCHAR(180) NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT fk_client_user FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE
);
```

```
CREATE TABLE IF NOT EXISTS admins (
    id INT AUTO_INCREMENT PRIMARY KEY,
    user_id INT NOT NULL UNIQUE,
    full_name VARCHAR(120) NOT NULL,
    surname VARCHAR(120) NOT NULL,
    employee_code VARCHAR(40) NULL UNIQUE,
    contact_info VARCHAR(180) NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT fk_admin_user FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE
);
```

```
CREATE TABLE IF NOT EXISTS financial_profiles (
    id INT AUTO_INCREMENT PRIMARY KEY,
    client_id INT NOT NULL UNIQUE,
    gross_salary DECIMAL(12,2) NOT NULL DEFAULT 0,
    deductions DECIMAL(12,2) NOT NULL DEFAULT 0,
    monthly_expenses DECIMAL(12,2) NOT NULL DEFAULT 0,
    current_savings DECIMAL(12,2) NOT NULL DEFAULT 0,
    net_salary DECIMAL(12,2) NOT NULL DEFAULT 0,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,
    CONSTRAINT fk_financial_client FOREIGN KEY (client_id) REFERENCES clients(id) ON DELETE CASCADE
);
```

```
CREATE TABLE IF NOT EXISTS banks (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(160) NOT NULL,
    contact_info VARCHAR(220) NOT NULL,
    website VARCHAR(255) NOT NULL,
    plan_type VARCHAR(80) NOT NULL,
    expected_return DECIMAL(5,2) NOT NULL,
    risk ENUM('Low', 'Medium', 'High') NOT NULL,
    liquidity ENUM('High', 'Medium', 'Low') NOT NULL,
    horizon ENUM('Short', 'Medium', 'Long') NOT NULL,
    allows_monthly TINYINT(1) NOT NULL DEFAULT 0,
    details TEXT NULL,
    created_by_admin_id INT NULL,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    CONSTRAINT fk_bank_admin FOREIGN KEY (created_by_admin_id) REFERENCES admins(id) ON DELETE SET NULL
);
```

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```
CREATE TABLE IF NOT EXISTS investment_plans (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  client_id INT NOT NULL,  
  bank_id INT NULL,  
  user_plan_name VARCHAR(160) NOT NULL,  
  investment_amount DECIMAL(12,2) NOT NULL,  
  monthly_contribution TINYINT(1) NOT NULL DEFAULT 0,  
  monthly_amount DECIMAL(12,2) NOT NULL DEFAULT 0,  
  investment_goal VARCHAR(255) NOT NULL,  
  risk ENUM('Low', 'Medium', 'High') NOT NULL,  
  liquidity ENUM('High', 'Medium', 'Low') NOT NULL,  
  horizon ENUM('Short', 'Medium', 'Long') NOT NULL,  
  expected_return DECIMAL(5,2) NOT NULL,  
  score INT NOT NULL DEFAULT 0,  
  bank_investment_url VARCHAR(255) NOT NULL,  
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,  
  CONSTRAINT fk_plan_client FOREIGN KEY (client_id) REFERENCES clients(id) ON DELETE  
  CASCADE,  
  CONSTRAINT fk_plan_bank FOREIGN KEY (bank_id) REFERENCES banks(id) ON DELETE SET NULL  
);
```

```
CREATE TABLE IF NOT EXISTS activity_logs (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  user_id INT NULL,  
  client_id INT NULL,  
  admin_id INT NULL,  
  action VARCHAR(80) NOT NULL,  
  entity_type VARCHAR(80) NOT NULL,  
  entity_id INT NULL,  
  description VARCHAR(255) NULL,  
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  CONSTRAINT fk_activity_user FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE SET NULL,  
  CONSTRAINT fk_activity_client FOREIGN KEY (client_id) REFERENCES clients(id) ON DELETE SET  
  NULL,  
  CONSTRAINT fk_activity_admin FOREIGN KEY (admin_id) REFERENCES admins(id) ON DELETE SET  
  NULL  
);
```

```
CREATE TABLE IF NOT EXISTS auth_otps (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  user_id INT NOT NULL,  
  purpose ENUM('login', 'password_reset') NOT NULL,  
  otp_hash VARCHAR(255) NOT NULL,  
  attempts INT NOT NULL DEFAULT 0,  
  expires_at DATETIME NOT NULL,  
  used_at DATETIME NULL,  
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  CONSTRAINT fk_auth_otp_user FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE,  
  INDEX idx_auth_otps_user_purpose (user_id, purpose, expires_at)  
);
```

```
INSERT INTO users (email, password_hash, role, status)  
VALUES ('admin@investsmart.local',
```

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```
'$2y$10$Slg7oTxQlhPf5ltaabyL7.LeVLwxfh6s1WHavQwwwthhhO959nQG.', 'admin', 'active')
ON DUPLICATE KEY UPDATE email = email;
```

```
INSERT INTO admins (user_id, full_name, surname, employee_code, contact_info)
SELECT u.id, 'System', 'Admin', 'ADM-001', 'admin@investsmart.local'
FROM users u
WHERE u.email = 'admin@investsmart.local'
AND NOT EXISTS (SELECT 1 FROM admins a WHERE a.user_id = u.id);
```

```
INSERT INTO banks (name, contact_info, website, plan_type, expected_return, risk, liquidity,
horizon, allows_monthly, details, created_by_admin_id) VALUES
('Standard Bank', 'investments@standardbank.co.za | 0860 123 000',
'https://www.standardbank.co.za/southafrica/personal/products-and-services/grow-your-
money/investment-solutions/digital-save-and-invest', 'Fixed Plan', 7.20, 'Low', 'Low',
'Short', 0, 'Fixed-term investment option for users who prefer capital stability and
predictable returns.', 1),
('FNB', 'invest@fnb.co.za | 087 575 9404', 'https://www.fnb.co.za/for-me/save-and-
invest.html', 'Flexi Plan', 8.35, 'Low', 'High', 'Short', 1, 'Flexible savings and
investment access with recurring contribution support.', 1),
('Nedbank', 'wealth@nedbank.co.za | 0800 555 111', 'https://personal.nedbank.co.za/private-
clients/invest/invest-with-us.html', 'Retirement/Income Plan', 9.40, 'Medium', 'Low',
'Long', 1, 'Long-term retirement and income-focused solution for balanced investors.', 1),
('Absa', 'investment@absa.co.za | 0860 111 515',
'https://www.absa.co.za/personal/investment-management/investment-accounts/', 'Growth Plan',
12.60, 'Medium', 'Medium', 'Medium', 1, 'Balanced growth investment for users seeking
moderate risk and medium-term returns.', 1),
('Capitec', 'invest@capitecbank.co.za | 0860 102 043',
'https://www.capitecbank.co.za/personal/save/', 'Flexi Plan', 8.75, 'Medium', 'High',
'Short', 1, 'Simple flexible investment structure with accessible monthly contribution
options for users who can accept moderate risk.', 1),
('Investec', 'wealth@investec.co.za | 0860 443 443',
'https://www.investec.com/en_za/focus/investing/how-to-save-and-invest.html', 'Equity Plan',
15.80, 'High', 'High', 'Long', 0, 'Higher-growth equity-linked investment option for users
with long horizons and higher risk tolerance.', 1)
ON DUPLICATE KEY UPDATE website = VALUES(website), contact_info = VALUES(contact_info),
details = VALUES(details), expected_return = VALUES(expected_return);
```

OTP Update Script for Existing Databases
USE investsmart;

```
CREATE TABLE IF NOT EXISTS auth_otps (
  id INT AUTO_INCREMENT PRIMARY KEY,
  user_id INT NOT NULL,
  purpose ENUM('login', 'password_reset') NOT NULL,
  otp_hash VARCHAR(255) NOT NULL,
  attempts INT NOT NULL DEFAULT 0,
  expires_at DATETIME NOT NULL,
  used_at DATETIME NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  CONSTRAINT fk_auth_otp_user FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE,
  INDEX idx_auth_otps_user_purpose (user_id, purpose, expires_at)
);
```